

SAVANT on every dataset

SAVANT at its 2026-08-07 defaults, budgeted with `--sa-steps N` so each instance’s budget comes from its own dimension, $\max(500000, 668n^{1.37})$. The reference solvers were then given *SAVANT’s own measured per-instance CPU time* on each set. None of them is matched at these budgets and none is expected to be: a process launch is a few milliseconds, HGS-CVRP builds a dense $n \times n$ matrix, LKH-3 generates a candidate set and AILS-II starts a JVM, all outside the time flag. Every solver’s *achieved* time is therefore printed next to its cost, and should be read first. Every run was checked by `tools/validate.py`, which recomputes each solution from the coordinates independently of the solver that produced it.

Linux-7.0.0-28-generic-x86_64-with-glibc2.39, 24 cores. The 10,000-instance sets ran on every core; X and XL ran single-threaded, because their tables report a per-instance time and `time_ms` is measured inside SAVANT’s parallel region, where it would otherwise absorb the contention between threads. Generated 2026-08-07 18:48.

1 Generated instances

10,000 instances per set, uniform points in the unit square, *continuous* distances. No reference solutions ship with them, so the cost stands on its own. LKH-3 has no float mode — EUC_2D is rounded by TSPLIB rules — so it cannot be run here without changing the problem, and AILS-II was not asked for on these sets. `min` and `max` are over the *instances*, not over repeated runs of one instance: they describe the spread of the set.

set	solver	<i>m</i>	mean cost	min	max	sd	vs SAVANT %	CPU ms/inst	× asked	—sa-steps	valid
n20	SAVANT	10,000	6.13043	3.73284	9.54073	0.80991	—	34.2	1.00	500,000	ok
	HGS	10,000	6.13020	3.73284	9.54073	0.80973	-0.004	34.9	1.02	—	ok
n50	SAVANT	10,000	10.41082	7.29155	15.95232	1.22499	—	34.9	1.00	500,000	ok
	HGS	10,000	10.38330	7.29155	15.96155	1.22296	-0.264	35.7	1.02	—	ok
n100	SAVANT	10,000	15.72705	11.83471	22.59039	1.80726	—	36.1	1.00	500,000	ok
	HGS	10,000	15.86787	11.86522	22.89857	1.84432	+0.895	37.2	1.03	—	ok

Table 1: Generated sets, at SAVANT’s own per-instance budget. ‘vs SAVANT %’ is the paired mean difference as a percentage of SAVANT’s mean; negative means the other solver is cheaper. The CPU column is what each solver actually spent.

2 XML100

10,000 instances of exactly 100 customers, *integer* distances (TSPLIB EUC_2D). Every one has a *proven optimum*, so the gap is a true optimality gap rather than a bound.

solver	returned	mean cost	min	max	sd	mean gap %	median	worst	at optimum	CPU ms/inst	× asked	valid
SAVANT	10,000	16109.32	3799	80340	9503.57	+0.859	+0.603	+13.251	866 (9.2%)	40.7	1.00	ok
HGS	9,910	16156.90	3831	80391	9520.46	+1.145	+1.036	+8.235	267 (2.9%)	41.7	1.03	ok
LKH	9,367	16763.54	3851	80743	10156.77	+4.068	+2.633	+61.915	597 (6.4%)	54.1	1.33	ok

Table 2: XML100 against the proven optima, at SAVANT’s per-instance budget. ‘returned’ is how many of the 10,000 that solver produced a feasible answer for; every other column is computed on the 9,364 instances *all* of them returned, so the rows are comparable. Averaging each solver over its own subset would reward whichever one failed on the hardest instances.

2.1 By generator attribute

The layout of Table 4 of the XL paper (arXiv:2601.11467): the mean optimality gap of every subgroup defined by one attribute value. The four blocks are four different partitions of the same 10,000 instances, so each one averages back to the overall row. The decoding of the instance name’s four digits was measured from the `.vrp` files, not assumed; `report_bench.py --verify` re-checks it.

attribute	level	m	SAVANT		HGS		LKH	
			gap %	opt %	gap %	opt %	gap %	opt %
Depot pos.	R	3,192	+0.874	8.5	+1.170	2.3	+4.171	5.6
	C	3,116	+0.764	13.0	+0.922	4.8	+4.261	10.3
	E	3,056	+0.942	6.2	+1.347	1.4	+3.764	3.1
Customer pos.	R	3,132	+0.846	11.2	+1.236	3.6	+4.326	6.3
	C	3,115	+0.875	7.9	+1.008	2.6	+3.655	7.0
	RC	3,117	+0.857	8.7	+1.190	2.3	+4.222	5.8
Demand dist.	U	1,410	+0.393	15.9	+0.634	3.4	+1.708	7.0
	1-10	1,397	+0.910	7.8	+1.344	2.1	+3.629	5.7
	5-10	1,405	+0.799	7.8	+1.193	2.7	+3.854	5.1
	1-100	1,385	+1.003	7.1	+1.344	2.8	+5.485	6.6
	50-100	1,392	+0.823	9.1	+1.164	2.8	+5.179	6.5
	Q	1,361	+0.979	8.2	+1.369	1.6	+5.376	5.4
	SL	1,014	+1.214	8.7	+0.917	5.1	+3.038	9.1
Route size r	VS	1,229	+0.971	2.9	+1.229	0.3	+8.082	0.0
	S	1,489	+0.998	2.3	+1.376	0.3	+6.723	0.0
	M	1,649	+1.027	3.9	+1.339	0.4	+4.617	0.1
	L	1,665	+0.836	7.1	+1.211	1.7	+3.354	0.8
	VL	1,666	+0.783	12.5	+1.071	4.7	+2.175	6.2
	UL	1,666	+0.588	24.2	+0.695	8.8	+0.798	28.8
	Overall	9,364	+0.859	9.2	+1.145	2.9	+4.068	6.4

Table 3: Mean optimality gap by generator attribute, and the share of the subgroup solved to optimality. Levels: depot Random / Central / Eccentric; customers Random / Clustered / Random-Clustered; demand Unitary, four sampling ranges, Quadrant-dependent, Small-with-a-few-Large; route size Very Short (3-5) through Ultra Long (25-50).

3 X, instance by instance

Integer distances. 61 of the 100 references are proven optima and the rest are best known solutions, so the gap bounds the optimality gap from above.

solver	returned	mean gap %	median	best	worst	own subset	CPU s/inst	× asked	valid
SAVANT	100	+1.375	+1.187	+0.066	+5.773	+1.466	0.15	1.00	ok
HGS	95	+2.812	+2.765	+0.047	+5.767	+2.948	0.16	1.03	ok
LKH	59	+13.375	+7.569	+0.071	+67.579	+13.375	0.28	1.85	ok
AILS	100	+0.604	+0.529	+0.000	+1.814	+0.713	2.31	15.02	ok

Table 4: X: the first block is computed on the 59 instances *every* solver returned, so the rows are comparable. ‘own subset’ is the same mean over whatever that solver alone managed — the difference between the two columns is how much a solver is flattered by the instances it failed on. ‘× asked’ is achieved CPU over the budget requested: read it before the gap columns, because a solver several times over budget is not being compared at equal cost.

Table 5: X, one row per instance. ‘s’ is that solver’s own measured time for that instance, single-core. The last column is the budget --sa-steps N gave SAVANT, i.e. $\max(500000, 668 n^{1.37})$.

instance	n	BKS	SAVANT			HGS			LKH			AILS			-sa-steps
			cost	gap %	s	cost	gap %	s	cost	gap %	s	cost	gap %	s	
X-n101-k25	100	27,591	28,039	+1.62	0.02	27,628	+0.13	0.16	—	—	—	27,591	+0.00	1.06	500,000
X-n106-k14	105	26,362	26,440	+0.30	0.02	26,585	+0.85	0.16	26,912	+2.09	0.17	26,370	+0.03	1.06	500,000
X-n110-k13	109	14,971	15,110	+0.93	0.03	15,011	+0.27	0.16	15,425	+3.03	0.17	14,971	+0.00	1.08	500,000
X-n115-k10	114	12,747	13,079	+2.60	0.03	12,753	+0.05	0.16	12,756	+0.07	0.17	12,750	+0.02	1.07	500,000
X-n120-k6	119	13,332	13,415	+0.62	0.04	13,472	+1.05	0.16	13,417	+0.64	0.17	13,367	+0.26	1.06	500,000
X-n125-k30	124	55,539	56,371	+1.50	0.02	56,214	+1.22	0.16	57,303	+3.18	0.20	55,564	+0.05	1.09	500,000
X-n129-k18	128	28,940	29,411	+1.63	0.02	29,408	+1.62	0.16	30,071	+3.91	0.17	29,110	+0.59	1.06	514,818
X-n134-k13	133	10,916	10,935	+0.17	0.03	11,104	+1.72	0.16	11,320	+3.70	0.17	10,949	+0.30	1.06	542,566
X-n139-k10	138	13,590	13,652	+0.46	0.03	13,674	+0.62	0.16	13,991	+2.95	0.17	13,602	+0.09	1.06	570,703
X-n143-k7	142	15,700	15,837	+0.87	0.03	15,943	+1.55	0.16	16,193	+3.14	0.17	15,726	+0.17	1.06	593,487
X-n148-k46	147	43,448	44,112	+1.53	0.02	44,183	+1.69	0.16	48,120	+10.75	0.18	43,563	+0.26	1.06	622,302
X-n153-k22	152	21,220	22,413	+5.62	0.04	21,630	+1.93	0.16	—	—	—	21,231	+0.05	1.06	651,481
X-n157-k13	156	16,876	16,916	+0.24	0.04	16,932	+0.33	0.15	17,037	+0.95	0.17	16,876	+0.00	1.06	675,082
X-n162-k11	161	14,138	14,197	+0.42	0.04	14,175	+0.26	0.16	14,259	+0.86	0.17	14,171	+0.23	1.09	704,900
X-n167-k10	166	20,557	20,803	+1.20	0.04	21,278	+3.51	0.16	21,099	+2.64	0.17	20,592	+0.17	1.06	735,062
X-n172-k51	171	45,607	46,093	+1.07	0.03	46,314	+1.55	0.16	—	—	—	45,769	+0.36	1.06	765,563
X-n176-k26	175	47,812	50,572	+5.77	0.04	49,254	+3.02	0.16	51,431	+7.57	0.17	47,954	+0.30	1.06	790,202
X-n181-k23	180	25,569	25,690	+0.47	0.05	25,740	+0.67	0.16	25,930	+1.41	0.17	25,591	+0.09	1.09	821,296
X-n186-k15	185	24,145	24,477	+1.38	0.04	24,725	+2.40	0.16	25,376	+5.10	0.18	24,231	+0.36	1.06	852,710
X-n190-k8	189	16,980	17,104	+0.73	0.06	17,450	+2.77	0.16	17,468	+2.87	0.18	17,020	+0.24	1.06	878,069
X-n195-k51	194	44,225	44,622	+0.90	0.04	45,017	+1.79	0.16	—	—	—	44,322	+0.22	1.07	910,049
X-n200-k36	199	58,578	59,863	+2.19	0.04	59,853	+2.18	0.16	62,856	+7.30	0.20	58,659	+0.14	1.07	942,334
X-n204-k19	203	19,565	20,094	+2.70	0.05	20,062	+2.54	0.16	20,279	+3.65	0.18	19,793	+1.17	1.06	968,380
X-n209-k16	208	30,656	30,751	+0.31	0.05	31,606	+3.10	0.16	32,118	+4.77	0.18	30,714	+0.19	1.07	1,001,205
X-n214-k11	213	10,856	11,064	+1.92	0.06	11,379	+4.82	0.16	13,776	+26.90	0.19	10,998	+1.31	1.06	1,034,323
X-n219-k73	218	117,595	117,673	+0.07	0.05	117,791	+0.17	0.16	120,308	+2.31	0.20	117,606	+0.01	1.07	1,067,731
X-n223-k34	222	40,437	40,917	+1.19	0.05	41,481	+1.58	0.16	43,169	+6.76	0.19	40,781	+0.85	1.06	1,094,661
X-n228-k23	227	25,742	26,403	+2.57	0.06	26,364	+2.42	0.16	27,171	+5.55	0.18	25,745	+0.01	1.06	1,128,578
X-n233-k16	232	19,230	19,407	+0.92	0.06	19,843	+3.19	0.15	27,071	+40.77	0.20	19,337	+0.56	1.07	1,162,773
X-n237-k14	236	27,042	27,251	+0.77	0.08	27,724	+2.52	0.16	28,256	+4.49	0.20	27,117	+0.28	1.07	1,190,325
X-n242-k48	241	82,751	83,713	+1.16	0.05	84,662	+2.31	0.16	90,853	+9.79	0.22	83,085	+0.40	1.07	1,225,010
X-n247-k50	246	37,274	38,983	+4.58	0.07	—	—	—	—	—	—	37,398	+0.33	1.07	1,259,962

instance	n	BKS	SAVANT				HGS				LKH				AILS				-sa-steps
			cost	gap	%	s	cost	gap	%	s	cost	gap	%	s	cost	gap	%	s	
X-n251-k28	250	38,684	39,045	+0.93	0.06	0.06	39,720	+2.68	0.16	0.16	40,976	+5.92	0.19	0.19	39,046	+0.94	1.07	1.07	1,288,113
X-n256-k16	255	18,839	18,949	+0.58	0.08	0.08	19,242	+2.14	0.16	0.16	25,431	+34.99	0.21	0.21	18,880	+0.22	1.07	1.07	1,323,538
X-n261-k13	260	26,558	27,015	+1.72	0.07	0.07	27,853	+4.88	0.16	0.16	28,354	+6.76	0.19	0.19	26,709	+0.57	1.07	1.07	1,359,220
X-n266-k58	265	75,478	76,319	+1.11	0.06	0.06	77,242	+2.34	0.16	0.16	82,952	+9.90	0.23	0.23	75,849	+0.49	1.07	1.07	1,395,157
X-n270-k35	269	35,291	35,580	+0.82	0.07	0.07	36,063	+2.19	0.16	0.16	50,212	+42.28	0.24	0.24	35,580	+0.82	1.09	1.09	1,424,088
X-n275-k28	274	21,245	21,441	+0.92	0.09	0.09	21,714	+2.21	0.16	0.16	21,870	+2.94	0.19	0.19	21,412	+0.79	1.07	1.07	1,460,477
X-n280-k17	279	33,503	34,732	+3.67	0.09	0.09	34,567	+3.18	0.16	0.16	35,748	+6.70	0.23	0.23	33,689	+0.56	1.07	1.07	1,497,111
X-n284-k15	283	20,215	20,788	+2.83	0.09	0.09	21,282	+5.28	0.16	0.16	21,811	+7.90	0.20	0.20	20,311	+0.47	1.08	1.08	1,526,595
X-n289-k60	288	95,151	96,707	+1.64	0.07	0.07	97,696	+2.67	0.16	0.16	—	—	—	—	95,589	+0.46	1.07	1.07	1,563,666
X-n294-k50	293	47,161	47,834	+1.43	0.07	0.07	48,461	+2.76	0.16	0.16	62,105	+31.69	0.21	0.21	47,324	+0.35	1.07	1.07	1,600,977
X-n298-k31	297	34,231	34,727	+1.45	0.08	0.08	35,796	+4.57	0.16	0.16	40,370	+17.93	0.20	0.20	34,677	+1.30	1.07	1.07	1,630,995
X-n303-k21	302	21,736	22,163	+1.96	0.09	0.09	22,337	+2.76	0.16	0.16	23,702	+9.04	0.20	0.20	21,981	+1.13	1.07	1.07	1,668,729
X-n308-k13	307	25,859	26,318	+1.78	0.12	0.12	26,754	+3.46	0.16	0.16	27,554	+6.55	0.20	0.20	26,124	+1.02	1.07	1.07	1,706,695
X-n313-k71	312	94,043	94,953	+0.97	0.07	0.07	96,443	+2.55	0.16	0.16	—	—	—	—	94,487	+0.47	1.07	1.07	1,744,890
X-n317-k53	316	78,355	78,731	+0.48	0.09	0.09	78,984	+0.80	0.16	0.16	—	—	—	—	78,782	+0.54	1.07	1.07	1,775,610
X-n322-k28	321	29,834	30,149	+1.06	0.10	0.10	30,904	+3.59	0.16	0.16	35,848	+20.16	0.21	0.21	29,927	+0.31	1.08	1.08	1,814,213
X-n327-k20	326	27,532	27,831	+1.09	0.11	0.11	28,834	+4.73	0.16	0.16	29,740	+8.02	0.21	0.21	27,763	+0.84	1.07	1.07	1,853,039
X-n331-k15	330	31,102	31,353	+0.81	0.15	0.15	32,230	+3.63	0.16	0.16	32,421	+4.24	0.21	0.21	31,321	+0.70	1.08	1.08	1,884,258
X-n336-k84	335	139,111	142,331	+2.31	0.08	0.08	143,144	+2.90	0.16	0.16	—	—	—	—	140,025	+0.66	1.07	1.07	1,923,480
X-n344-k43	343	42,050	42,550	+1.19	0.10	0.10	43,454	+3.34	0.16	0.16	49,216	+17.04	0.25	0.25	42,473	+1.01	1.08	1.08	1,986,686
X-n351-k40	350	25,896	26,338	+1.71	0.09	0.09	27,010	+4.30	0.16	0.16	35,138	+35.69	0.26	0.26	26,278	+1.48	1.12	1.12	2,042,441
X-n359-k29	358	51,505	52,154	+1.26	0.11	0.11	53,777	+4.41	0.16	0.16	57,089	+10.84	0.22	0.22	51,823	+0.62	1.07	1.07	2,106,668
X-n367-k17	366	22,814	23,754	+4.12	0.14	0.14	23,666	+3.73	0.16	0.16	24,700	+8.27	0.22	0.22	23,112	+1.31	1.08	1.08	2,171,428
X-n376-k94	375	147,713	148,105	+0.27	0.10	0.10	148,642	+0.63	0.16	0.16	—	—	—	—	147,841	+0.09	1.10	1.10	2,244,911
X-n384-k52	383	65,928	66,644	+1.09	0.11	0.11	67,748	+2.76	0.16	0.16	88,966	+34.94	0.25	0.25	66,479	+0.84	1.08	1.08	2,310,780
X-n393-k38	392	38,260	38,664	+1.06	0.13	0.13	39,658	+3.65	0.16	0.16	44,184	+15.48	0.26	0.26	38,596	+0.88	1.08	1.08	2,385,494
X-n401-k29	400	66,154	66,806	+0.99	0.14	0.14	68,391	+3.38	0.16	0.16	72,902	+10.20	0.23	0.23	66,411	+0.39	1.09	1.09	2,452,441
X-n411-k19	410	19,712	19,869	+0.80	0.16	0.16	20,573	+4.37	0.16	0.16	21,262	+7.86	0.24	0.24	19,919	+1.05	1.07	1.07	2,536,824
X-n420-k130	419	107,798	108,873	+1.00	0.10	0.10	110,588	+2.59	0.16	0.16	—	—	—	—	108,258	+0.43	1.08	1.08	2,613,422
X-n429-k61	428	65,449	66,198	+1.14	0.13	0.13	67,680	+3.41	0.16	0.16	101,117	+54.50	0.25	0.25	65,978	+0.81	1.08	1.08	2,690,632
X-n439-k37	438	36,391	36,618	+0.62	0.18	0.18	37,517	+3.09	0.16	0.16	—	—	—	—	36,623	+0.64	1.08	1.08	2,777,128
X-n449-k29	448	55,233	56,521	+2.33	0.14	0.14	57,917	+4.86	0.16	0.16	—	—	—	—	56,052	+1.48	1.12	1.12	2,864,358
X-n459-k26	458	24,139	24,583	+1.84	0.18	0.18	25,531	+5.77	0.16	0.16	29,332	+21.51	0.26	0.26	24,577	+1.81	1.08	1.08	2,952,311
X-n469-k138	468	221,824	225,396	+1.61	0.12	0.12	226,218	+1.98	0.16	0.16	—	—	—	—	223,326	+0.68	1.10	1.10	3,040,978
X-n480-k70	479	89,449	90,657	+1.35	0.15	0.15	92,219	+3.10	0.16	0.16	109,310	+22.20	0.28	0.28	89,922	+0.53	1.08	1.08	3,139,323
X-n491-k59	490	66,483	67,471	+1.49	0.15	0.15	69,376	+4.35	0.16	0.16	91,941	+38.29	0.28	0.28	67,363	+1.32	1.10	1.10	3,238,508
X-n502-k39	501	69,226	69,382	+0.23	0.23	0.23	70,025	+1.15	0.16	0.16	72,025	+4.04	0.28	0.28	69,391	+0.24	1.09	1.09	3,338,521
X-n513-k21	512	24,201	24,576	+1.55	0.23	0.23	25,139	+3.88	0.16	0.16	27,620	+14.13	0.28	0.28	24,454	+1.05	1.11	1.11	3,439,349
X-n524-k153	523	154,593	159,354	+3.08	0.18	0.18	—	—	—	—	—	—	—	—	155,156	+0.36	1.13	1.13	3,540,982
X-n536-k96	535	94,846	96,041	+1.26	0.16	0.16	97,844	+3.16	0.16	0.16	—	—	—	—	95,431	+0.62	1.11	1.11	3,652,760
X-n548-k50	547	86,700	87,213	+0.59	0.24	0.24	88,447	+2.01	0.16	0.16	—	—	—	—	86,989	+0.33	1.10	1.10	3,765,469
X-n561-k42	560	42,717	43,308	+1.38	0.20	0.20	44,831	+4.95	0.16	0.16	54,286	+27.08	0.31	0.31	43,157	+1.03	1.12	1.12	3,888,607
X-n573-k30	572	50,673	51,326	+1.29	0.28	0.28	52,255	+3.12	0.17	0.17	—	—	—	—	51,166	+0.97	1.11	1.11	4,003,215
X-n586-k159	585	190,316	192,672	+1.24	0.17	0.17	195,016	+2.47	0.16	0.16	—	—	—	—	191,264	+0.50	1.15	1.15	4,128,383
X-n599-k92	598	108,451	109,625	+1.08	0.19	0.19	112,178	+3.44	0.16	0.16	—	—	—	—	109,431	+0.90	1.13	1.13	4,254,583
X-n613-k62	612	59,535	60,645	+1.86	0.21	0.21	62,615	+5.17	0.16	0.16	99,768	+67.58	0.34	0.34	60,390	+1.44	1.13	1.13	4,391,631
X-n627-k43	626	62,164	62,756	+0.95	0.27	0.27	64,928	+4.45	0.17	0.17	—	—	—	—	62,731	+0.91	1.15	1.15	4,529,844
X-n641-k35	640	63,682	64,519	+1.31	0.28	0.28	66,488	+4.41	0.16	0.16	74,668	+17.25	0.35	0.35	64,491	+1.27	1.14	1.14	4,669,205
X-n655-k131	654	106,780	107,027	+0.23	0.24	0.24	107,594	+0.76	0.16	0.16	—	—	—	—	106,951	+0.16	1.15	1.15	4,809,699
X-n670-k130	669	146,332	153,894	+5.17	0.28	0.28	—	—	—	—	—	—	—	—	148,662	+1.59	1.12	1.12	4,961,468
X-n685-k75	684	68,205	69,308	+1.62	0.27	0.27	71,145	+4.31	0.16	0.16	—	—	—	—	68,929	+1.06	1.18	1.18	5,114,501
X-n701-k44	700	81,923	82,922	+1.22	0.29	0.29	85,743	+4.66	0.18	0.18	—	—	—	—	83,002	+1.32	1.15	1.15	5,279,110
X-n716-k35	715	43,373	44,121	+1.72	0.33	0.33	46,224	+6.57	0.17	0.17	—	—	—	—	43,935	+1.30	1.14	1.14	5,434,702
X-n733-k159	732	136,187	137,475	+0.95	0.24	0.24	141,298	+3.75	0.16	0.16	—	—	—	—	136,984	+0.59	1.16	1.16	5,612,503
X-n749-k98	748	77,269	78,495	+1.59	0.27	0.27	80,440	+4.10	0.16	0.16	—	—	—	—	78,343	+1.39	1.16	1.16	5,781,248
X-n766-k71	765	114,417	116,399	+1.73	0.35	0.35	—	—	—	—	—	—	—	—	116,091	+1.46	1.16	1.16	5,962,009
X-n783-k48	782	72,386	73,419	+1.43	0.36	0.36	76,137	+5.18	0.17	0.17	—	—	—	—	73,493	+1.53	1.17	1.17	6,144,261
X-n801-k40	800	73,305	73,587	+0.38	0.51	0.51	76,644	+4.55	0.17	0.17	—	—	—	—	74,058	+1.03	1.20	1.20	6,338,839
X-n819-k171	818	158,121	160,313	+1.39	0.29	0.29	162,920	+3.04	0.17	0.17	—	—	—	—	159,515	+0.88	1.17	1.17	6,535,043
X-n837-k142	836	193,737	195,316	+0.82	0.31	0.31	199,356	+2.90	0.16	0.16	—	—	—	—	195,083	+0.69	1.18	1.18	6,732,851
X-n856-k95	855	88,965	89,420	+0.51	0.43	0.43	90,882	+2.15	0.16	0.16	—	—	—	—	89,328	+0.41	1.18	1.18	6,943,365
X-n876-k59	875	99,299	100,																

solver	returned	mean gap %	median	best	worst	own subset	CPU s/inst	× asked	valid
SAVANT	100	+1.336	+1.243	+0.066	+3.909	+1.411	5.77	1.00	ok
HGS	95	+4.886	+4.090	+0.289	+11.717	+4.886	6.19	1.07	ok
LKH [†]	18	+20.338	+10.214	+2.063	+89.590	+20.338	182.74	31.65	ok
AILS	100	+2.296	+1.619	+0.094	+11.082	+2.352	18.59	3.22	ok

Table 6: XL: the first block is computed on the 95 instances *every* solver returned, so the rows are comparable. ‘own subset’ is the same mean over whatever that solver alone managed — the difference between the two columns is how much a solver is flattered by the instances it failed on. ‘× asked’ is achieved CPU over the budget requested: read it before the gap columns, because a solver several times over budget is not being compared at equal cost. [†] returned under 50% of the set, so it does not define the common subset — its row is its own instances only and is not comparable with the others.

Table 7: XL, one row per instance. ‘s’ is that solver’s own measured time for that instance, single-core. The last column is the budget --sa-steps N gave SAVANT, i.e. $\max(500000, 668 n^{1.37})$.

instance	n	SAVANT				HGS			LKH			AILS			-sa-
		BKS	cost	gap %	s	cost	gap %	s	cost	gap %	s	cost	gap %	s	
XL-n1048-k237	1,047	380,107	398,099	+4.73	0.53	—	—	—	—	—	—	384,385	+1.13	6.28	9,16
XL-n1094-k157	1,093	112,431	112,780	+0.31	0.56	113,776	+1.20	5.79	114,751	+2.06	6.47	112,673	+0.22	6.28	9,72
XL-n1141-k112	1,140	95,638	96,968	+1.39	0.56	98,538	+3.03	5.79	157,937	+65.14	6.93	96,446	+0.84	6.23	10,29
XL-n1188-k96	1,187	104,402	105,915	+1.45	0.62	107,543	+3.01	5.79	132,075	+26.51	6.47	105,592	+1.14	6.25	10,88
XL-n1234-k55	1,233	96,603	97,903	+1.35	0.71	102,719	+6.33	5.82	104,642	+8.32	6.46	97,495	+0.92	6.28	11,46
XL-n1281-k29	1,280	31,883	31,810	+2.34	1.08	33,036	+6.28	5.79	32,997	+6.16	6.52	31,681	+1.92	6.29	12,06
XL-n1328-k19	1,327	38,201	38,884	+1.79	1.72	40,865	+6.97	5.82	40,267	+5.41	6.60	38,933	+1.92	6.28	12,67
XL-n1374-k278	1,373	232,940	236,204	+1.40	0.62	239,509	+2.82	5.79	—	—	—	234,667	+0.74	6.33	13,28
XL-n1421-k232	1,420	384,646	390,990	+1.65	0.67	393,431	+2.28	5.79	—	—	—	387,209	+0.67	6.31	13,91
XL-n1468-k151	1,467	250,028	252,391	+0.95	0.76	257,327	+2.92	5.79	—	—	—	252,271	+0.90	6.33	14,54
XL-n1514-k106	1,513	92,398	93,304	+0.98	0.91	96,524	+4.47	5.79	109,489	+18.50	6.84	93,228	+0.90	6.31	15,17
XL-n1561-k75	1,560	101,540	102,122	+0.57	1.36	103,863	+2.29	5.84	104,976	+3.38	6.92	101,989	+0.44	6.31	15,82
XL-n1608-k39	1,607	48,009	49,018	+2.10	1.35	51,342	+6.94	5.84	50,798	+5.81	6.96	48,553	+1.13	6.34	16,48
XL-n1654-k11	1,653	36,370	36,766	+1.09	3.17	39,221	+7.84	5.92	38,163	+4.93	6.99	37,262	+2.45	6.37	17,13
XL-n1701-k562	1,700	520,843	529,366	+1.64	0.70	534,331	+2.59	5.81	—	—	—	524,840	+0.77	6.37	17,80
XL-n1748-k271	1,747	173,759	176,318	+1.47	0.91	180,800	+4.05	5.84	—	—	—	175,179	+0.82	6.40	18,48
XL-n1794-k163	1,793	141,724	142,332	+0.43	1.24	145,311	+2.53	5.80	—	—	—	142,387	+0.47	6.40	19,15
XL-n1841-k126	1,840	213,981	217,532	+1.66	1.28	223,853	+4.61	5.80	251,644	+17.60	7.44	215,771	+0.84	6.38	19,84
XL-n1888-k82	1,887	143,558	145,180	+1.13	1.43	151,838	+5.77	5.95	—	—	—	144,904	+0.94	6.40	20,53
XL-n1934-k46	1,933	52,980	53,963	+1.86	1.84	56,378	+6.41	5.82	72,768	+37.35	7.54	53,770	+1.49	6.44	21,22
XL-n1981-k31	1,980	32,546	33,447	+2.77	3.76	35,292	+8.44	5.81	35,604	+9.40	7.57	33,162	+1.89	6.41	21,93
XL-n2028-k617	2,027	544,166	552,855	+1.60	0.96	563,557	+3.56	5.80	—	—	—	548,224	+0.75	6.47	22,65
XL-n2074-k264	2,073	421,250	426,211	+1.18	1.12	436,694	+3.67	5.86	—	—	—	426,007	+1.13	6.50	23,36
XL-n2121-k186	2,120	283,105	285,633	+0.89	1.26	294,543	+4.04	5.87	—	—	—	284,735	+0.58	6.55	24,09
XL-n2168-k138	2,167	127,210	128,805	+1.25	1.54	133,898	+5.26	5.84	—	—	—	129,142	+1.52	6.48	24,82
XL-n2214-k131	2,213	154,652	155,373	+0.47	2.04	158,309	+2.36	5.81	—	—	—	155,397	+0.48	6.50	25,55
XL-n2261-k54	2,260	98,826	100,716	+1.91	2.47	106,132	+7.39	5.85	113,064	+14.41	8.13	100,510	+1.70	6.53	26,29
XL-n2307-k34	2,306	47,936	48,980	+2.18	3.05	52,042	+8.57	5.87	53,083	+10.74	8.29	49,012	+2.24	6.55	27,03
XL-n2354-k631	2,353	940,634	949,433	+0.94	1.21	962,010	+2.27	5.82	—	—	—	943,457	+0.30	6.58	27,79
XL-n2401-k408	2,400	463,084	468,838	+1.24	1.39	479,755	+3.60	5.84	—	—	—	468,063	+1.08	6.60	28,55
XL-n2447-k290	2,446	218,633	225,422	+3.11	1.87	231,944	+6.09	5.91	—	—	—	221,114	+1.13	6.59	29,30
XL-n2494-k194	2,493	361,057	366,452	+1.49	1.74	369,612	+2.37	5.88	—	—	—	363,038	+0.55	6.59	30,08
XL-n2541-k121	2,540	146,345	147,225	+0.60	2.60	152,378	+4.12	5.83	—	—	—	147,453	+0.76	6.62	30,86
XL-n2587-k66	2,586	73,254	74,546	+1.76	3.01	78,596	+7.29	5.91	96,036	+31.10	8.95	75,057	+2.46	6.65	31,62
XL-n2634-k17	2,633	31,619	32,536	+2.90	5.91	34,968	+10.59	5.98	34,683	+9.69	9.10	33,043	+4.50	6.64	32,41
XL-n2681-k540	2,680	797,906	809,910	+1.50	1.58	820,514	+2.83	5.90	—	—	—	808,627	+1.34	6.69	33,21
XL-n2727-k546	2,726	431,108	431,768	+0.15	1.83	434,764	+0.85	5.85	—	—	—	431,702	+0.14	6.72	33,99
XL-n2774-k286	2,773	407,617	411,634	+0.99	1.91	417,715	+2.48	5.90	—	—	—	411,483	+0.95	6.68	34,80
XL-n2821-k208	2,820	216,622	220,629	+1.85	2.35	229,278	+5.84	5.85	—	—	—	219,263	+1.22	6.71	35,61
XL-n2867-k120	2,866	165,916	167,673	+1.06	2.89	176,072	+6.12	6.29	—	—	—	167,763	+1.11	6.92	36,41
XL-n2914-k95	2,913	88,902	90,500	+1.80	3.37	95,249	+7.14	5.85	168,549	+89.59	9.84	91,047	+2.41	6.79	37,23
XL-n2961-k55	2,960	107,945	109,901	+1.81	2.95	117,854	+9.18	6.23	—	—	—	110,089	+1.99	6.75	38,05
XL-n3007-k658	3,006	522,126	526,465	+0.83	1.77	538,148	+3.07	5.85	—	—	—	525,485	+0.64	6.85	38,87
XL-n3054-k461	3,053	782,194	790,028	+1.00	1.97	799,840	+2.26	5.85	—	—	—	790,870	+1.11	6.85	39,70
XL-n3101-k311	3,100	245,766	252,744	+2.84	2.76	258,548	+5.20	5.97	—	—	—	248,101	+0.95	6.89	40,54
XL-n3147-k232	3,146	256,417	258,387	+0.77	2.53	266,055	+3.76	5.89	—	—	—	258,947	+0.99	6.84	41,37
XL-n3194-k161	3,193	148,596	150,461	+1.26	2.95	157,440	+5.95	5.98	—	—	—	151,389	+1.88	6.86	42,22
XL-n3241-k115	3,240	221,249	224,055	+1.27	2.98	230,322	+4.10	6.05	—	—	—	223,050	+0.81	6.88	43,07
XL-n3287-k30	3,286	40,190	40,785	+1.48	10.38	43,355	+7.88	5.84	—	—	—	41,748	+3.88	6.91	43,91
XL-n3334-k934	3,333	1,452,353	1,463,117	+0.74	2.01	1,478,629	+1.81	5.95	—	—	—	1,462,944	+0.73	6.98	44,77
XL-n3408-k524	3,407	678,024	685,313	+1.08	2.38	699,515	+3.17	5.90	—	—	—	686,866	+1.30	6.98	46,14
XL-n3484-k436	3,483	703,319	704,470	+0.16	3.10	707,468	+0.59	5.92	—	—	—	704,344	+0.15	7.00	47,56
XL-n3561-k229	3,560	209,191	212,275	+1.47	2.92	220,141	+5.23	5.87	—	—	—	212,665	+1.66	7.04	49,00
XL-n3640-k211	3,639	189,531	191,537	+1.06	3.30	199,641	+5.33	5.92	—	—	—	192,067	+1.34	7.11	50,50
XL-n3721-k77	3,720	162,630	166,454	+2.35	5.24	174,194	+7.11	6.39	—	—	—	165,263	+1.62	7.14	52,04
XL-n3804-k29	3,803	52,838	53,555	+1.36	10.05	57,919	+9.62	6.20	—	—	—	54,779	+3.67	7.18	53,64
XL-n3888-k1010	3,887	1,877,119	1,967,544	+4.82	3.48	—	—	—	—	—	—	1,962,446	+4.55	11.51	55,27
XL-n3975-k687	3,974	525,560	530,255	+0.89	2.79	543,132	+3.34	5.99	—	—	—	530,511	+0.94	7.31	56,97
XL-n4063-k347	4,062	548,330	553,735	+0.99	3.60	567,410	+3.48	6.04	—	—	—	558,980	+1.94	7.36	58,71
XL-n4153-k291	4,152	355,780	360,141	+1.23	3.62	367,240	+3.22	5.92	—	—	—	359,225	+0.97	7.40	60,50
XL-n4245-k203	4,244	229,521	230,683	+0.51	5.45	238,063	+3.72	6.18	—	—	—	232,024	+1.09	7.47	62,34
XL-n4340-k148	4,339	243,895	246,578	+1.10	5.23	260,675	+6.88	6.29	—	—	—	251,352	+3.06	7.54	64,26
XL-n4436-k48	4,435	61,431	62,293	+1.40	12.18	66,189	+7.75	6.35	—	—	—	63,381	+3.17	7.81	66,22
XL-n4535-k1134	4,534	1,203,512	1,204,309	+0.07	3.68	1,206,992	+0.29	5.96	—	—	—	1,204,642	+0.09	7.73	68,25
XL-n4635-k790	4,634	609,734	617,074	+1.20	3.45	632,871	+3.79	5.88	—	—	—	661,873	+8.55	8.43	70,32
XL-n4738-k487	4,737	759,649	767,118	+0.98	4.16	778,644	+3.16	6.14	—	—	—	778,253	+2.45	7.79	72,47
XL-n4844-k321	4,843	404,429	413,790	+2.31	5.71	429,354	+6.16	6.39	—	—	—	411,039	+1.63	7.85	74,71
XL-n4951-k203	4,950	284,769	287,809	+1.07	6.14	303,413	+6.55	6.60	—	—	—	292,657	+2.77	8.50	76,98

instance	n	BKS	SAVANT			HGS			LKH			AILS			-sa-
			cost	gap %	s	cost	gap %	s	cost	gap %	s	cost	gap %	s	
XL-n5061-k184	5,060	161,480	163,646	+1.34	6.72	170,981	+5.88	6.08	—	—	—	163,932	+1.52	8.29	79,33
XL-n5174-k55	5,173	61,254	62,958	+2.78	12.09	68,366	+11.61	6.06	—	—	—	64,414	+5.16	8.06	81,77
XL-n5288-k1246	5,287	1,958,205	1,983,961	+1.32	3.90	2,007,530	+2.52	6.16	—	—	—	2,028,843	+3.61	32.62	84,25
XL-n5406-k783	5,405	1,040,141	1,049,206	+0.87	4.67	1,064,647	+2.36	5.95	—	—	—	1,058,348	+1.75	8.31	86,83
XL-n5526-k553	5,525	336,843	337,808	+0.29	6.18	342,699	+1.74	5.92	—	—	—	338,814	+0.59	8.36	89,48
XL-n5649-k401	5,648	644,562	658,687	+2.19	6.87	678,241	+5.23	6.69	—	—	—	657,302	+1.98	8.49	92,22
XL-n5774-k290	5,773	249,895	253,362	+1.39	6.32	265,589	+6.28	6.07	—	—	—	256,310	+2.57	8.68	95,03
XL-n5902-k122	5,901	217,012	220,227	+1.48	10.41	239,858	+10.53	7.89	—	—	—	225,158	+3.75	8.73	97,93
XL-n6034-k61	6,033	64,251	65,764	+2.35	20.75	70,863	+10.29	7.09	—	—	—	67,891	+5.67	8.73	100,95
XL-n6168-k1922	6,167	1,527,390	1,549,930	+1.48	4.77	—	—	—	—	—	—	1,620,106	+6.07	133.02	104,03
XL-n6305-k1042	6,304	1,176,002	1,188,463	+1.06	6.01	1,214,217	+3.25	6.07	—	—	—	1,241,964	+5.61	22.50	107,21
XL-n6445-k628	6,444	996,117	1,001,433	+0.53	6.47	1,017,725	+2.17	6.53	—	—	—	1,028,619	+3.26	9.18	110,48
XL-n6588-k473	6,587	333,822	337,198	+1.01	7.08	349,832	+4.80	6.05	—	—	—	339,839	+1.80	9.33	113,86
XL-n6734-k330	6,733	447,486	451,197	+0.83	8.56	465,789	+4.09	7.25	—	—	—	459,627	+2.71	9.44	117,33
XL-n6884-k148	6,883	181,585	185,974	+2.42	14.52	198,842	+9.50	7.09	—	—	—	186,792	+2.87	10.14	120,92
XL-n7037-k38	7,036	70,760	71,451	+0.98	34.32	78,421	+10.83	6.50	—	—	—	76,955	+8.75	10.53	124,62
XL-n7193-k1683	7,192	2,958,205	2,974,099	+0.54	6.51	3,011,254	+1.79	6.18	—	—	—	3,050,946	+3.14	51.98	128,42
XL-n7353-k1471	7,352	1,537,702	1,538,856	+0.08	7.62	1,545,095	+0.48	6.02	—	—	—	1,550,326	+0.82	17.09	132,35
XL-n7516-k859	7,515	572,984	581,220	+1.44	7.99	593,308	+3.55	5.98	—	—	—	602,632	+5.17	18.09	136,39
XL-n7683-k602	7,682	701,151	706,544	+0.77	9.14	721,463	+2.90	6.42	—	—	—	733,377	+4.60	13.19	140,56
XL-n7854-k365	7,853	659,075	669,757	+1.62	14.17	—	—	—	—	—	—	665,607	+0.99	10.53	144,86
XL-n8028-k294	8,027	266,295	269,475	+1.19	13.38	286,251	+7.49	6.30	—	—	—	276,754	+3.93	11.36	149,28
XL-n8207-k108	8,206	118,050	122,665	+3.91	17.00	126,390	+7.06	7.86	—	—	—	122,462	+3.74	12.64	153,86
XL-n8389-k2028	8,388	3,353,305	3,405,273	+1.55	8.21	—	—	—	—	—	—	3,498,105	+4.32	239.09	158,55
XL-n8575-k1297	8,574	1,088,376	1,095,768	+0.68	9.02	1,122,875	+3.17	6.10	—	—	—	1,158,278	+6.42	25.09	163,39
XL-n8766-k1032	8,765	905,060	911,604	+0.72	9.75	930,537	+2.81	6.61	—	—	—	951,711	+5.15	31.87	168,40
XL-n8960-k634	8,959	772,906	778,835	+0.77	13.12	794,071	+2.74	6.66	—	—	—	832,784	+7.75	15.14	173,52
XL-n9160-k379	9,159	323,596	329,845	+1.93	15.96	350,749	+8.39	7.10	—	—	—	338,869	+4.72	12.49	178,85
XL-n9363-k209	9,362	205,369	208,343	+1.45	28.17	221,115	+7.67	6.51	—	—	—	213,905	+4.16	13.25	184,31
XL-n9571-k55	9,570	106,344	108,804	+2.31	48.13	118,804	+11.72	10.86	—	—	—	118,129	+11.08	13.39	189,94
XL-n9784-k2774	9,783	4,076,936	4,112,517	+0.87	10.80	4,138,550	+1.51	8.03	—	—	—	4,229,496	+3.74	246.10	195,75
XL-n10001-k1570	10,000	2,330,666	2,354,409	+1.02	12.16	2,376,311	+1.96	7.16	—	—	—	2,446,851	+4.99	149.43	201,73